EC304 — Network Theory

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Module 4 Exam Notes: Network Functions, Two-Port Networks, Resonance & Filters

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Complex Frequency and Network Functions

The concept of complex frequency generalizes real sinusoidal and exponential signals into a unified mathematical framework. It forms the foundation for characterizing linear time-invariant (LTI) networks in the complex frequency s-plane.

1.1 Concept of Complex Frequency

Definition of Complex Frequency

A generalized damped sinusoidal signal is expressed as:

$$f(t) = A\,e^{\sigma t}\,\cos(\omega t + \theta) = \text{Re}\left\{A\,e^{j\theta}\,e^{(\sigma+j\omega)t}\right\} = \text{Re}\left\{\mathbf{F}\,e^{st}\right\} \tag{1}$$

The parameter $s = \sigma + j\omega$ is the **complex frequency** (measured in s^{-1} or complex nepers/radians per second):

- $\sigma = \text{Re}(s)$ is the **Neper frequency** (damping or growth rate in Np/s).
- $\omega = \text{Im}(s)$ is the **radian angular frequency** (oscillation rate in rad/s).

Special Cases of the Complex Frequency Variable $s = \sigma + j\omega$

1. $\sigma = 0, \omega = 0 \implies s = 0$: Constant DC signal ($f(t) = A$).
2. $\sigma \neq 0, \omega = 0 \implies s = \sigma$: Pure exponential signal ($e^{\sigma t}$; decaying if $\sigma < 0$, growing if $\sigma > 0$).
3. $\sigma = 0, \omega \neq 0 \implies s = j\omega$: Undamped steady-state sinusoidal signal ($A\cos(\omega t)$).
4. $\sigma < 0, \omega \neq 0$: Damped sinusoidal signal (exponentially decaying oscillatory transient).
5. $\sigma > 0, \omega \neq 0$: Growing sinusoidal signal (unstable active system).

1.2 Definition and Classification of Network Functions

Network Function Definition

For a linear, lumped, time-invariant network with all initial conditions set to zero (relaxed state), a **Network Function** $H(s)$ is defined as the ratio of the Laplace transform of the output response $R(s)$ to the Laplace transform of the input excitation $E(s)$:

$$H(s) = \frac{R(s)}{E(s)} \iff R(s) = H(s) \cdot E(s) \tag{2}$$

Network functions are broadly classified into two categories:

1. **Driving-Point Immittance Functions:** Excitation and response are measured at the **same port**:

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- *Driving-point impedance:* $Z_{in}(s) = \dfrac{V_1(s)}{I_1(s)}$
- *Driving-point admittance:* $Y_{in}(s) = \dfrac{I_1(s)}{V_1(s)} = \dfrac{1}{Z_{in}(s)}$

2. **Transfer Functions:** Excitation and response are measured at **different ports**:

- *Voltage Transfer Function (Gain):* $G_{21}(s) = \dfrac{V_2(s)}{V_1(s)}$
- *Current Transfer Function (Gain):* $\alpha_{21}(s) = \dfrac{I_2(s)}{I_1(s)}$
- *Transfer Impedance:* $Z_{21}(s) = \dfrac{V_2(s)}{I_1(s)}$
- *Transfer Admittance:* $Y_{21}(s) = \dfrac{I_2(s)}{V_1(s)}$

2. Poles and Zeros of Immittance Functions and their Properties

Any network function of a lumped LTI circuit can be expressed as a rational function of s with real coefficients:

$$H(s) = \frac{N(s)}{D(s)} = \frac{a_m s^m + a_{m-1}s^{m-1} + \cdots + a_1 s + a_0}{b_n s^n + b_{n-1}s^{n-1} + \cdots + b_1 s + b_0} = K\frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{k=1}^{n}(s - p_k)} \tag{3}$$

2.1 Definitions of Poles and Zeros

Poles and Zeros

- **Zeros (z_i):** The values of complex frequency s for which $H(s) = 0$ (roots of the numerator $N(s) = 0$). At a transmission zero, the network completely suppresses the input signal.
- **Poles (p_k):** The values of complex frequency s for which $H(s) \to \infty$ (roots of the denominator $D(s) = 0$). Poles define the natural frequencies (modes) and determine stability.
- **Critical Frequencies at Infinity:** If $n > m$, $H(s)$ has $(n-m)$ zeros at $s = \infty$. If $m > n$, $H(s)$ has $(m - n)$ poles at $s = \infty$. The total number of zeros equals the total number of poles in the extended complex plane.

2.2 Properties of Driving-Point Immittances (Positive Real Functions)

For any passive, bilateral electrical network without internal active sources, the driving-point immittance $Z(s)$ or $Y(s)$ must be a **Positive Real (PR) Function** (Brune's condition):

$$\mathrm{Re}\{Z(s)\} \geq 0 \quad \text{whenever} \quad \mathrm{Re}(s) \geq 0 \tag{4}$$

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Necessary and Sufficient Properties of Passive Immittance Functions

1. **Hurwitz Denominator and Numerator:** All poles and zeros of $Z(s)$ must lie in the left half of the s-plane ($\text{Re}(s) \leq 0$). They cannot reside in the open right-half plane (RHP).
2. **Imaginary Axis Poles and Zeros:** Any poles and zeros located on the $j\omega$-axis must be **simple** (multiplicity of 1). Multiple poles on the $j\omega$-axis are strictly forbidden because they generate unbounded polynomial-exponential terms ($t \sin \omega t$).
3. **Residue Condition:** The residues of $Z(s)$ at simple poles on the $j\omega$-axis must be **real and strictly positive**.
4. **Degree Constraint:** The degrees of the numerator polynomial $N(s)$ and denominator polynomial $D(s)$ can differ at most by 1: $|m - n| \leq 1$.
5. **Lowest Degree Constraint:** The lowest power of s in $N(s)$ and $D(s)$ can differ at most by 1.
6. **Foster Alternation Law (LC Immittance):** For purely reactive lossless LC networks, all poles and zeros lie exclusively on the $j\omega$-axis, are simple, and must strictly **interlace (alternate)**:
$$0 \leq \omega_{z1} < \omega_{p1} < \omega_{z2} < \omega_{p2} < \cdots \tag{5}$$
Furthermore, the slope of reactance with respect to frequency is strictly positive: $\frac{dX(\omega)}{d\omega} > 0$.

3. Sinusoidal Steady-State Response from Pole-Zero Locations

The steady-state frequency response $H(j\omega)$ of any network function can be evaluated directly by geometric inspection of its pole-zero constellation in the s-plane.

3.1 Geometric Vector Formulation

Evaluating $H(s)$ on the imaginary axis $s = j\omega$:

$$H(j\omega) = K \frac{(j\omega - z_1)(j\omega - z_2)\cdots(j\omega - z_m)}{(j\omega - p_1)(j\omega - p_2)\cdots(j\omega - p_n)} \tag{6}$$

Each term $(j\omega - z_i)$ represents a directed complex vector in the s-plane from the zero z_i to the test frequency point $s = j\omega$:

$$(j\omega - z_i) = M_i\, e^{j\psi_i}, \quad \text{where } M_i = |j\omega - z_i|, \quad \psi_i = \angle(j\omega - z_i) \tag{7}$$

Similarly, each term $(j\omega - p_k)$ represents a vector from the pole p_k to the test point $j\omega$:

$$(j\omega - p_k) = N_k\, e^{j\theta_k}, \quad \text{where } N_k = |j\omega - p_k|, \quad \theta_k = \angle(j\omega - p_k) \tag{8}$$

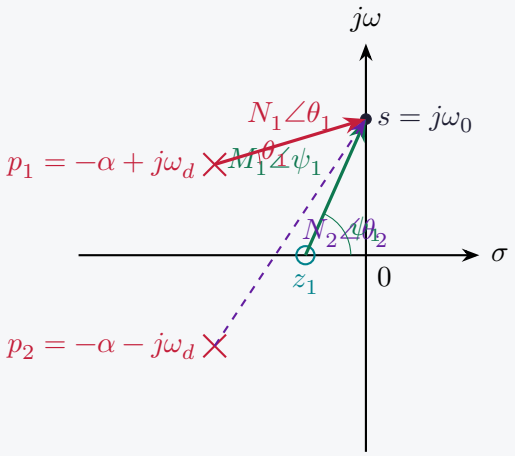
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Magnitude and Phase Computation via Pole-Zero Geometry

$$|H(j\omega)| = |K| \cdot \frac{M_1 M_2 \cdots M_m}{N_1 N_2 \cdots N_n} = |K| \cdot \frac{\prod(\text{distances from zeros to } j\omega)}{\prod(\text{distances from poles to } j\omega)} \tag{9}$$

$$\angle H(j\omega) = \angle K + \sum_{i=1}^{m} \psi_i - \sum_{k=1}^{n} \theta_k = \angle K + \sum(\text{zero angles}) - \sum(\text{pole angles}) \tag{10}$$

TikZ Schematic: Geometric Vector Evaluation in the Complex s-Plane



4. The Convolution Theorem in Electrical Networks

The convolution theorem establishes the fundamental mathematical bridge between time-domain input-output relationships and s-domain transfer function multiplication.

4.1 Mathematical Definition and Circuit Interpretation

Time Convolution Theorem

Let $h(t)$ be the **impulse response** of a relaxed LTI network (the response when excited by $x(t) = \delta(t)$). The zero-state response $y(t)$ to an arbitrary continuous excitation $x(t)$ applied at $t = 0$ is given by the convolution integral:

$$y(t) = x(t) * h(t) = \int_{0^-}^{t} x(\tau)\, h(t-\tau)\, d\tau = \int_{0^-}^{t} h(\tau)\, x(t-\tau)\, d\tau \tag{11}$$

In the transformed frequency domain:

$$\mathcal{L}\{x(t) * h(t)\} = X(s) \cdot H(s) \iff Y(s) = H(s)X(s) \tag{12}$$

where $H(s) = \mathcal{L}\{h(t)\}$ is the transfer function of the network.

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Relation Between Impulse Response and Step Response

If $s(t)$ represents the unit step response ($x(t) = u(t)$) of a linear network:

$$s(t) = u(t) * h(t) = \int_{0^-}^{t} h(\tau)\, d\tau \iff h(t) = \frac{ds(t)}{dt} \tag{13}$$

The unit impulse response is the exact time derivative of the unit step response.

5. Two-Port Network Parameters and Interconnections

A two-port network is a four-terminal electrical network having two distinct access ports: Port 1 (input port, terminals $1-1'$) and Port 2 (output port, terminals $2-2'$). Standard sign convention assumes terminal currents I_1 and I_2 enter the positive terminal of their respective ports.

TikZ Schematic: Standard Terminal Variable Convention for a Two-Port Network



5.1 The Six Fundamental Two-Port Parameter Sets

Definitions and Equations of Two-Port Parameter Sets

1. **Impedance (Z) Parameters (Open-Circuit Parameters):**

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \implies \begin{aligned} V_1 &= z_{11} I_1 + z_{12} I_2 \\ V_2 &= z_{21} I_1 + z_{22} I_2 \end{aligned} \tag{14}$$

$z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0}$ (input impedance), $z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0}$ (reverse transfer impedance),
$z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0}$ (forward transfer impedance), $z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0}$ (output impedance).

2. **Admittance (Y) Parameters (Short-Circuit Parameters):**

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \implies \begin{aligned} I_1 &= y_{11} V_1 + y_{12} V_2 \\ I_2 &= y_{21} V_1 + y_{22} V_2 \end{aligned} \tag{15}$$

$[Y] = [Z]^{-1} \implies y_{11} = \frac{z_{22}}{\Delta z}, \quad y_{12} = -\frac{z_{12}}{\Delta z}, \quad y_{21} = -\frac{z_{21}}{\Delta z}, \quad y_{22} = \frac{z_{11}}{\Delta z}.$

3. **Transmission ($ABCD$) Parameters (Cascade / Chain Parameters):** Relates input terminal variables to output terminal variables under reverse current load

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convention:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \implies \begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned} \tag{16}$$

$A = \frac{V_1}{V_2}\big|_{I_2=0}$ (reverse voltage ratio), $B = \frac{V_1}{-I_2}\big|_{V_2=0}$ (transfer impedance),
$C = \frac{I_1}{V_2}\big|_{I_2=0}$ (transfer admittance), $D = \frac{I_1}{-I_2}\big|_{V_2=0}$ (reverse current ratio).

4. **Hybrid (h) Parameters:** Used extensively in small-signal bipolar transistor modeling:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} \implies \begin{aligned} V_1 &= h_{11}I_1 + h_{12}V_2 \\ I_2 &= h_{21}I_1 + h_{22}V_2 \end{aligned} \tag{17}$$

5. **Inverse Hybrid (g) Parameters:** Used in field-effect transistor modeling:

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} \implies \begin{aligned} I_1 &= g_{11}V_1 + g_{12}I_2 \\ V_2 &= g_{21}V_1 + g_{22}I_2 \end{aligned} \tag{18}$$

5.2 Reciprocity and Symmetry Conditions

Parameter Set	Reciprocity Condition	Symmetry Condition
Impedance (Z)	$z_{12} = z_{21}$	$z_{11} = z_{22}$
Admittance (Y)	$y_{12} = y_{21}$	$y_{11} = y_{22}$
Transmission ($ABCD$)	$AD - BC = 1$	$A = D$
Hybrid (h)	$h_{12} = -h_{21}$	$\Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$
Inverse Hybrid (g)	$g_{12} = -g_{21}$	$\Delta g = g_{11}g_{22} - g_{12}g_{21} = 1$

5.3 Interconnections of Two-Port Networks

When two-port networks N_A and N_B are interconnected, their overall parameter matrices combine algebraically:

Combination Rules for Interconnected Two-Ports

1. **Series-Series Connection:** Input ports connected in series, output ports in series:

$$[Z] = [Z_A] + [Z_B] \tag{19}$$

2. **Parallel-Parallel Connection:** Input ports in parallel, output ports in parallel:

$$[Y] = [Y_A] + [Y_B] \tag{20}$$

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3. **Cascade (Tandem) Connection:** Output of N_A connected to input of N_B:

$$[T] = [T_A] \times [T_B] = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \tag{21}$$

4. **Series-Parallel Connection:** Input ports in series, output ports in parallel:

$$[h] = [h_A] + [h_B] \tag{22}$$

Brune's Test for Validity: When combining two-port networks in series or parallel, circulating loop currents between the two networks can violate the fundamental port condition ($i_{\text{in}} = i_{\text{return}}$). Brune's test confirms validity: if the terminal voltage across the interconnection points is zero before connection, the matrices can be added directly without ideal isolation transformers.

6. Behaviors of Series and Parallel Resonant Circuits

Resonance is a phenomenon in reactive circuits where inductive and capacitive reactances cancel each other out at a specific frequency, causing the total input impedance to become purely real and the circuit power factor to equal unity ($\cos\phi = 1$).

6.1 Series Resonance

Consider an RLC branch connected in series across an AC source **V**. The total complex impedance is:

$$\mathbf{Z}(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right) \tag{23}$$

Series Resonance Equations

1. **Resonance Condition:** Net reactance is identically zero:

$$X_L = X_C \implies \omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz} \tag{24}$$

2. **Impedance at Resonance:** Minimum and purely resistive:

$$\mathbf{Z}(j\omega_0) = Z_{\min} = R \implies \mathbf{I}_0 = \frac{\mathbf{V}}{R} \quad \text{(Maximum Current)} \tag{25}$$

3. **Quality Factor (Q):** Defined as $2\pi \times \frac{\text{Maximum Energy Stored}}{\text{Energy Dissipated per Cycle}}$:

$$Q_0 = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}} \tag{26}$$

4. **Voltage Magnification:** Reactive voltages across L and C can far exceed source voltage:

$$V_L(j\omega_0) = V_C(j\omega_0) = Q_0 \cdot V \tag{27}$$

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5. **Half-Power Frequencies and Bandwidth (BW):** The frequencies ω_1 (lower) and ω_2 (upper) where current magnitude drops to $I_0/\sqrt{2}$ (70.7%) and dissipated active power drops to $P_{\max}/2$:

$$\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \implies BW = \Delta\omega = \omega_2 - \omega_1 = \frac{R}{L} = \frac{\omega_0}{Q_0} \tag{28}$$

Geometric center frequency relation: $\omega_0 = \sqrt{\omega_1\omega_2}$. For high-$Q$ circuits ($Q \geq 10$), arithmetic symmetry holds: $\omega_1 \approx \omega_0 - \frac{BW}{2}$, $\omega_2 \approx \omega_0 + \frac{BW}{2}$.

6.2 Parallel Resonance (Practical Tank Circuit)

A practical inductor possesses winding resistance R_L. The practical parallel tank consists of a coil (R_L in series with L) connected in parallel with an ideal capacitor C.

$$\mathbf{Y} = \frac{1}{R_L + j\omega L} + j\omega C = \frac{R_L - j\omega L}{R_L^2 + \omega^2 L^2} + j\omega C = \frac{R_L}{R_L^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R_L^2 + \omega^2 L^2}\right) \tag{29}$$

Practical Tank Circuit Resonance Conditions

1. **Unity Power Factor Resonant Frequency ω_r:** Setting net susceptance to zero:

$$\omega_r C = \frac{\omega_r L}{R_L^2 + \omega_r^2 L^2} \implies \omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}} = \omega_0\sqrt{1 - \frac{R_L^2 C}{L}} \tag{30}$$

Physical condition for parallel resonance to occur: $\dfrac{L}{C} > R_L^2$.

2. **Dynamic Impedance (Z_d):** At resonance, the circuit exhibits **maximum impedance** (anti-resonance):

$$Z_d = \frac{1}{Y_{\min}} = \frac{R_L^2 + \omega_r^2 L^2}{R_L} = \frac{L}{CR_L} \quad (\Omega) \tag{31}$$

Total line current is minimum: $I_{\min} = \frac{V}{Z_d} = \frac{VCR_L}{L}$.

3. **Current Magnification:** Circulating loop current between coil and capacitor is Q times line current:

$$I_{\text{circ}} \approx Q \cdot I_{\text{line}}, \quad \text{where } Q = \frac{\omega_r L}{R_L} \tag{32}$$

7. Introduction to Passive Electric Filters

An electric filter is a frequency-selective two-port network designed to transmit signals within specified frequency bands (**pass bands**) with minimal attenuation, while significantly attenuating signals outside those bands (**attenuation / stop bands**).

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7.1 Propagation Characteristics and Cut-Off Frequency

For a symmetrical reactive network terminated in its characteristic impedance Z_0:

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{\mathbf{I}_1}{\mathbf{I}_2} = e^{\gamma} = e^{\alpha+j\beta} \implies \gamma = \alpha + j\beta \tag{33}$$

where γ is the complex propagation constant, α is the attenuation constant (in Nepers), and β is the phase shift constant (in radians).

- **Pass Band:** $\alpha = 0$, signals pass with zero attenuation ($|\mathbf{V}_2/\mathbf{V}_1| = 1$), experiencing only a phase delay β.
- **Stop Band:** $\alpha > 0$, signal amplitude decays exponentially.
- **Cut-Off Frequency (f_c):** The boundary frequency demarcating the pass band from the stop band, defined where $\alpha = 0$ begins to increase.

7.2 Constant-k Prototype Filter Sections

A symmetrical ladder network has series impedance Z_1 and shunt impedance Z_2. It is termed a **constant-k filter** if:

$$Z_1 \cdot Z_2 = k^2 = R_0^2 \quad \text{(independent of frequency)} \tag{34}$$

where $R_0 = \sqrt{L/C}$ is the nominal design characteristic resistance.

Filter Type	Series Arm Z_1	Shunt Arm Z_2	Cut-off Frequencies & Design Formulas
Low-Pass (LPF)	**Inductor** L ($j\omega L$)	**Capacitor** C ($\frac{1}{j\omega C}$)	$f_c = \frac{1}{\pi\sqrt{LC}}$, $L = \frac{R_0}{\pi f_c}$, $C = \frac{1}{\pi f_c R_0}$
High-Pass (HPF)	**Capacitor** C ($\frac{1}{j\omega C}$)	**Inductor** L ($j\omega L$)	$f_c = \frac{1}{4\pi\sqrt{LC}}$, $L = \frac{R_0}{4\pi f_c}$, $C = \frac{1}{4\pi f_c R_0}$
Band-Pass (BPF)	**Series resonant** (L_1, C_1)	**Parallel resonant** (L_2, C_2)	$f_0 = \sqrt{f_1 f_2}$, $BW = f_2 - f_1$
Band-Reject (BRF)	**Parallel resonant** (L_1, C_1)	**Series resonant** (L_2, C_2)	Stops band between f_1 and f_2; passes DC and high AC.

Limitations of Constant-k Filters and Evolution to m-Derived Filters

1. **Gradual Attenuation Rise:** The attenuation α of a constant-k filter rises slowly beyond the cut-off frequency f_c.
2. **Mismatched Terminal Impedance:** The characteristic impedance $Z_{0T}(f)$ or $Z_{0\pi}(f)$ varies widely across the pass band and drops to zero or infinity at f_c, creating severe reflection losses when terminated in a fixed resistance R_0.
3. **m-Derived Filter Fix:** By incorporating a tuning parameter $m = \sqrt{1-(f_c/f_\infty)^2}$ ($0 < m \leq 1$), an infinite attenuation notch (f_∞) is introduced near the cut-off frequency,

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producing an extraordinarily sharp transition and a nearly constant characteristic impedance over 85% of the pass band.

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Quick Revision — Module 4

Network Functions, Two-Port Networks, Resonance and Filters | EC304

Key Formulas and Network Relations at a Glance

Complex Frequency	$s = \sigma + j\omega \quad (\sigma : \text{Np/s},\ \omega : \text{rad/s})$
Magnitude from Pole-Zero	$\lvert H(j\omega)\rvert = \lvert K\rvert \frac{\prod \lvert j\omega - z_i\rvert}{\prod \lvert j\omega - p_k\rvert} = \lvert K\rvert \frac{\text{Distance product from zeros}}{\text{Distance product from poles}}$
Phase from Pole-Zero	$\angle H(j\omega) = \sum \angle(j\omega - z_i) - \sum \angle(j\omega - p_k)$
Z **to** Y **Matrix Inverse**	$[Y] = [Z]^{-1}, \quad y_{11} = \frac{z_{22}}{\Delta z},\ y_{12} = -\frac{z_{12}}{\Delta z},\ y_{22} = \frac{z_{11}}{\Delta z}$
$ABCD$ **Transmission Cascade**	$[T_{\text{total}}] = [T_A] \times [T_B], \quad AD - BC = 1$ (Reciprocity)
Series Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Z_{\min} = R, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$
Bandwidth & Selectivity	$BW = \omega_2 - \omega_1 = \frac{R}{L} = \frac{\omega_0}{Q}, \quad \omega_0 = \sqrt{\omega_1 \omega_2}$
Parallel Dynamic Impedance	$\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}}, \quad Z_d = \frac{L}{CR_L}$ (Max Impedance)
Constant-k LPF Cut-off	$f_c = \frac{1}{\pi\sqrt{LC}}, \quad L = \frac{R_0}{\pi f_c}, \quad C = \frac{1}{\pi f_c R_0}$

One-Line Definitions

- **Complex Frequency (s):** Variable $s = \sigma + j\omega$ combining exponential damping rate and oscillation frequency.
- **Driving-Point Immittance:** The input impedance or admittance measured across a single terminal pair with internal relaxed state.
- **Transfer Function:** Ratio of response transform at one port to excitation transform at another port under zero initial states.
- **Hurwitz Polynomial:** A polynomial having all its roots located strictly in the closed left half of the complex s-plane.
- **Positive Real (PR) Function:** A rational function $Z(s)$ whose real part is non-negative whenever the real part of s is non-negative.
- **Impulse Response $h(t)$:** Time-domain output of an initially relaxed network when driven by an ideal Dirac delta excitation.
- **Quality Factor (Q):** Dimensionless figure of merit measuring energy storage efficiency relative to per-cycle dissipation.
- **Anti-Resonance:** Parallel resonance state exhibiting maximum terminal impedance and minimum source line current.
- **Cut-Off Frequency (f_c):** Frequency demarcating pass band ($\alpha = 0$) from stop band ($\alpha > 0$) in a filter network.
- **m-Derived Filter:** Modified ladder section with a resonance trap providing sharp attenuation notch and flat Z_0.

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Frequently Asked / Viva Questions

Q1. What are the necessary conditions for a rational function to be a Positive Real (PR) function?

$Z(s)$ must have real coefficients; all poles and zeros must lie in the left half s-plane; $j\omega$-axis poles must be simple with real, positive residues; and degrees of $N(s)$ and $D(s)$ can differ at most by 1.

Q2. Why must poles and zeros alternate on the imaginary axis for an LC driving-point immittance?

By Foster's Reactance Theorem, $\frac{dX(\omega)}{d\omega} > 0$ strictly holds for lossless LC networks. Reactance increases monotonically with frequency, forcing every zero ($X = 0$) to be succeeded by a pole ($X \to \infty$).

Q3. State the condition for reciprocity and symmetry in terms of $ABCD$ parameters.

Reciprocity holds if the transmission determinant is unity: $AD - BC = 1$. The network is symmetrical (electrically identical when ports are interchanged) if $A = D$.

Q4. Explain why series resonance gives 'voltage magnification' and parallel resonance 'current magnification'.

At series resonance, current is maximum, producing reactive voltages across L and C equal to Q times input voltage ($V_L = V_C = QV$). At parallel resonance, circulating current between L and C equals Q times line current ($I_{\text{circ}} = QI$).

Q5. What is Brune's Test, and why is it conducted before interconnecting two-port networks?

Brune's test verifies whether interconnecting two-ports creates circulating loop currents that violate the port condition ($I_{\text{in}} = I_{\text{out}}$). If open-circuit ground potential is non-zero, an ideal $1:1$ isolation transformer is required.

Q6. How does winding resistance R_L affect practical parallel tank resonant frequency?

Winding resistance lowers resonant frequency below the ideal value: $\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}} = \omega_0\sqrt{1 - \frac{R_L^2 C}{L}}$. If $R_L \geq \sqrt{L/C}$, resonance cannot occur.

Q7. State the relationship between resonant frequency ω_0 and half-power frequencies in series resonance.

ω_0 is the exact geometric mean of half-power frequencies: $\omega_0 = \sqrt{\omega_1\omega_2}$. For high-$Q$ circuits ($Q \geq 10$), it is also approximately their arithmetic mean: $\omega_0 \approx (\omega_1 + \omega_2)/2$.

Q8. Why does a constant-k filter have undesirable impedance characteristics in its pass band?

Its characteristic impedance $Z_{0T}(f) = R_0\sqrt{1 - (f/f_c)^2}$ drops to zero at f_c, while $Z_{0\pi}(f)$ approaches infinity. This impedance mismatch causes large reflection losses near cut-off when terminated in R_0.

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Frequently Asked / Viva Questions (contd.)

Q9. What is the physical meaning of poles in a network transfer function?

Poles represent the complex natural frequencies (modes) of the system. The real part dictates exponential decay or growth rate, while the imaginary part specifies unforced transient oscillation frequency.

Q10. State the relationship between convolution in time and transfer function in frequency domain.

In the time domain, response is the convolution of input with the impulse response: $y(t) = x(t) * h(t)$. In the s-domain, this integral simplifies into algebraic multiplication: $Y(s) = H(s) \cdot X(s)$.

Summary Matrix: Comprehensive Comparison of Two-Port Parameters and Resonant Circuits

Parameter / Circuit	Key Governing Relations	Reciprocity / Symmetry	Circuit Application
Z-Parameters	$V_1 = z_{11}I_1 + z_{12}I_2, \quad V_2 = z_{21}I_1 + z_{22}I_2$	$z_{12} = z_{21}, \quad z_{11} = z_{22}$	
Y-Parameters	$I_1 = y_{11}V_1 + y_{12}V_2, \quad I_2 = y_{21}V_1 + y_{22}V_2$	$y_{12} = y_{21}, \quad y_{11} = y_{22}$	
$ABCD$-Parameters	$V_1 = AV_2 - BI_2, \quad I_1 = CV_2 - DI_2$	$AD - BC = 1, \quad A = D$	
h-Parameters	$V_1 = h_{11}I_1 + h_{12}V_2, \quad I_2 = h_{21}I_1 + h_{22}V_2$	$h_{12} = -h_{21}, \quad \Delta h = 1$	
Series Resonant Circuit	$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Z_{\min} = R, \quad Q = \frac{\omega_0 L}{R}$	$V_L = V_C = QV$ **(Resonance)**	
Parallel Resonant Tank	$\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}}, \quad Z_d = \frac{L}{CR_L}$	$I_{\text{circ}} = QI_{\text{line}}$ **(Anti-resonance)**	

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